

Name → Arpit Anand

UID → 23BCSI2710

Section → KRA-1B

(8)

Date _____
Page _____

Q Given three integers n , a and b return n^{th} magical no. Since the ans may be very large return $10^9 + 7$.

A positive integer is magical if the number is divisible by either a or b .

$n=1, b=3, a=2$
or

$b=$

$$\frac{x}{a} + \frac{x}{b} = \frac{x}{\text{lcm}(a,b)}$$

Algorithm

We will first find $\text{lcm}(a, b)$

Then set $\text{low} = \min(a, b)$

Then set $\text{high} = n \times \min(a, b)$

while $\text{low} < \text{high}$

$$\text{mid} = (\text{low} + \text{high}) / 2$$

Then compute

$$\text{count} = \frac{\text{mid}}{a} + \frac{\text{mid}}{b} - \frac{\text{mid}}{\text{lcm}}$$

if $\text{count} < n$

then $\text{low} = \text{mid} + 1$

else

$\text{high} = \text{mid}$

Return $\text{low} \% (10^9 + 7)$

Dry run

$n=5, a=2, b=3$ $\text{lcm}=6$

x	$x/2$	$x/3$	$x/6$	count
6	3	2	1	4
8	4	2	1	5

so, answer = 8

22

66

15

815

15

215

Time complexity : $O(\log(n \times \min(a, b)))$
space : $O(1)$

Code

```
int magicalNo(int n, int a, int b) {
    const long long MOD = 1e9 + 7;
    long long g = gcd(a, b);
    long long lcm = (1LL * a * b) / g;
    long long low = min(a, b);
    long long high = 1LL * n * min(a, b);
    while (low < high) {
        long long mid = low + (high - low) / 2;
        long long count = mid / a + mid / b - mid / lcm;

        if (count < n) {
            low = mid + 1;
        }
        else {
            high = mid;
        }
    }
    return low % MOD;
}
```